INCH-POUND

MIL-C-50652/4A (AR)
12 December 1990
SUPERSEDING
MIL-C-50652/4(MU)
15 April 1971

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548046

(COUNTER, RANGE)

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range counter used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-50652 - Circuit Card Assemblies, General Specifications for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

F10548046	- Range Counter
F10548009	- PRF Counter

SPECIAL PACKAGING INSTRUCTION

PS10548046	- Packaging of Range Counter

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 General specifications. The assembly shall meet the requirements of MIL-C-50652.

3.2.1 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548046, all associated drawings, and with all requirements specified in applicable specifications and standards. The range counter shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. As specified in MIL-C-50652.

3.4. Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the card assembly.

3.5 Performance. For the requirements of 3.5.4 through 3.5.7, logic input and output levels shall be as follows:

Logic 0: 0.0 ± 0.5 volt direct current (Vdc) level.
Logic 1: + 3.5 ± 1.0 Vdc level.

3.5.1 Input power. The input power applied between terminals P1-30, P1-31 and P1-32 (+5 V) and terminals P1-24, 25 and 26 (+5 V return) shall be 5.0 ± 0.3 Vdc, at 200 milliamperes (mA), regulated; with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Clock signals. With a 7.5 ± 0.5 megahertz (MHz) square wave or sine wave applied to P1-43 there shall be a 7.5 MHz clock signal at P1-45 and P1-47. The input signal at P1-43 shall have an amplitude of 5.0 ± 0.2 volts into a 50-ohm load. The clock signal at P1-47 (7.5 MHz Cp) shall have an amplitude of 3.4 ± 1.0 Vdc and a period of 133 ± 10 nanoseconds (ns). The delayed clock signal at P1-45 (7.5 MHz pd) shall have an amplitude of 3.5 ± 1.0 Vdc, a period of 133 ± 10 ns, and be delayed 20ns + 20ns/-10ns from the signal at P1-47.

3.5.3 Input signals. The input signals to the range counter shall be the signals PRF RESET, PRF, T1, T1, T2, T3 and STROBE as shown in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ± 0.5 V to +3.5 ± 1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ± 1.0 V to 0.0 ± 1.0 V and return to +3.5 V.

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3.5.4 MIN RANGE INHIBIT output. With an input logic 0 level applied to P1-1 there shall be logic 1 level present at P1-14. With an input logic 1 level applied to P1-1 the MIN RANGE INHIBIT output signal as shown in Figure 1 shall be present at P1-14.

3.5.5 E/L COURSE output. With a logic 1 applied to P1-4 and the input logic levels applied sequentially as specified in Table I, the E/L course output signals at P1-51 shall be shown in Figure 1 with positive pulse durations as specified in Table I. With a logic "0" applied to P1-4, the E/L course output signal at P1-51 shall be a logic "0".

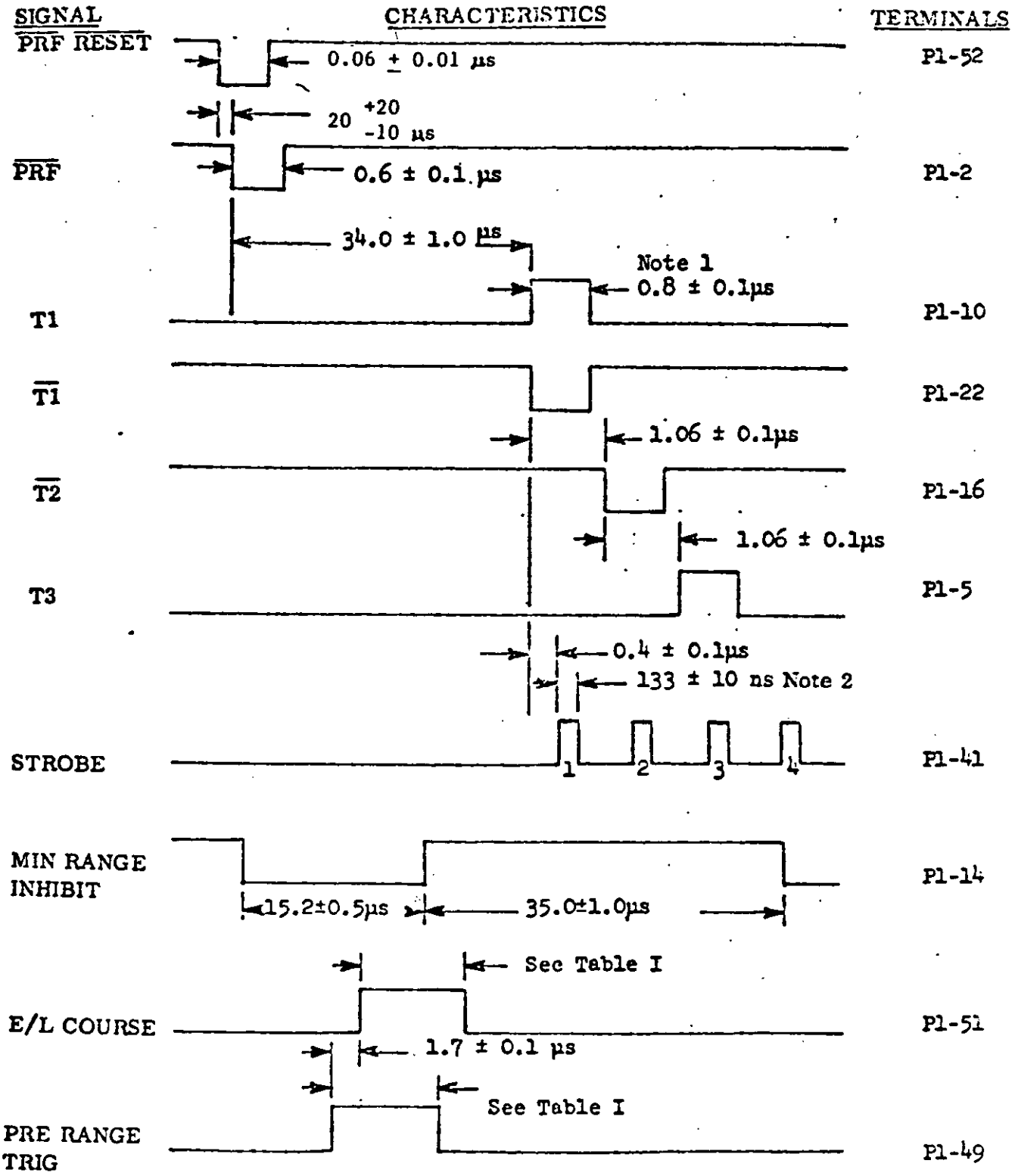
TABLE I. OUTPUT SIGNALS

INPUT TERMINALS									OUTPUT SIGNAL DURATIONS	
STEP	12	56	21	33	35	38	40	48	55	51 TERMINALS	49
1	1	0	1	0	0	0	0	0	1	0.27±0.05 us	2.65±0.15 us
2	0	0	1	0	0	0	0	0	1	0.40±0.05 us	2.79±0.20 us
3	1	1	1	0	0	0	0	0	1	0.53±0.08 us	2.92±0.25 us
4	1	0	0	0	0	0	0	0	1	0.80±0.10 us	3.19±0.20 us
5	1	0	1	1	0	0	0	0	1	1.33±0.15 us	3.72±0.20 us
6	1	0	1	0	1	0	0	0	1	2.40±0.20 us	4.79±0.35 us
7	1	0	1	0	0	1	0	0	1	4.54±0.60 us	6.92±0.35 us
8	1	0	1	0	0	0	1	0	1	8.81±0.58 us	11.19±0.8 us
9	1	0	1	0	0	0	0	1	1	17.35±1.0 us	19.73±0.8 us

3.5.6 PRE RANGE TRIG output. With the input logic levels applied sequentially as specified in Table I, the PRE RANGE TRIG output signal at P1-49 shall be as shown in Figure 1 with pulse duration as specified in Table I. There shall be a delay of 1.7 ± 0.1 microsecond (us) between the leading edge of the PRE RANGE TRIG output signal at P1-49 and the leading edge of the E/L COURSE output signal at P1-51 for all steps of Table I.

3.5.7 T2 output. At P1-46 the T2 output signal shall be present. The T2 signal shall be complementary to and have the same characteristics as T2 shown in Figure 1.

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NOTES: 1. All T pulses 0.8 ±0.1 µs wide.
2. STROBE signal is a train of 15 pulses each 133 ±10 ns wide.
3. All signals at a 20-kHz PRF.

FIGURE 1. INPUT AND OUTPUT SIGNALS

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification of the range counter.

4.2 First article. As specified in MIL-C-50652.

4.3 Inspection provisions. As specified in MIL-C-50652.

4.4 Special sampling. As specified in MIL-C-50652.

4.5 Inspection equipment. As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. As specified in MIL-C-50652.

4.5.3 Test equipment. In conjunction with 4.5 the test equipment listed in Table II shall be utilized in the performance of the applicable tests as specified in 4.6.3.

4.6 Methods of Inspection.

4.6.1 Environmental. As specified in MIL-C-50652.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

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TABLE II. TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ± 0.3 Vdc, regulated. Ac ripple: less than 50 mV peak-to-peak Capacity: 400 mA.
2. Signal generator	7.5 ± 0.5 MHz square wave or sine wave. Amplitude: 5.0 ± 0.2 V peak-to-peak into 50 ohms.
3. Oscilloscope	Dual channel: Rise time of scope plus measuring probe; 10 ns or less. Sweep period: 50 ns per cm or greater. Input resistance: 8 megohms or greater.
4. Test fixture	As specified in Figure 2. See Table III for test fixture parts list.

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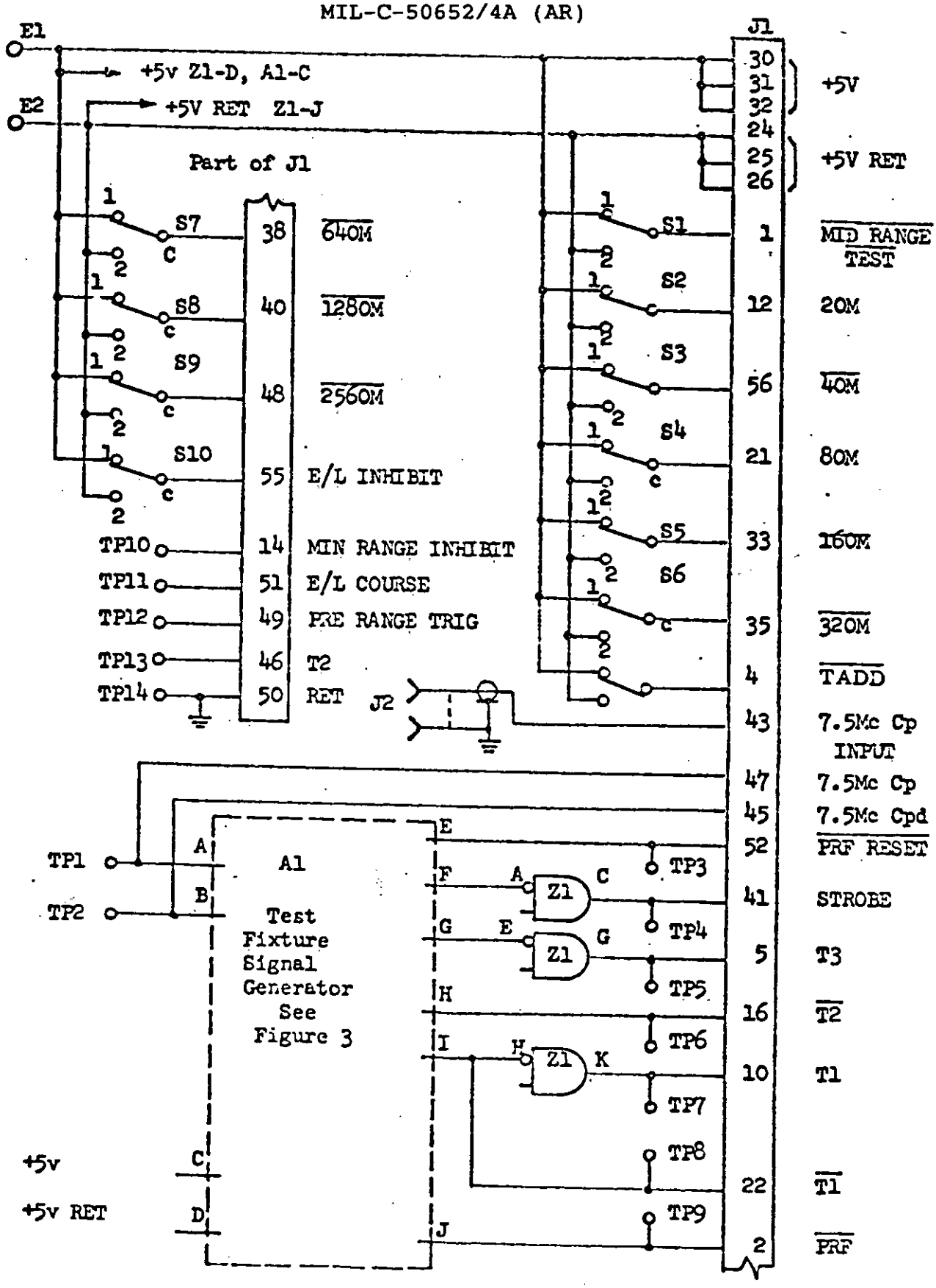


FIGURE 2. TEST FIXTURE SETUP

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TABLE III. TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
A1	Test fixture signal generator in accordance with Figure 3, and Table IV or applicable circuits on a PRF counter assembly, F10548009.
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector: Any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, PA
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
S1 through S11	Switch SPDT.
TP1 thru TP14	Tip jack, H. H. Smith 1501 or equivalent.
Z1	Integrated circuit, SG-140-03 Sylvania or equivalent.

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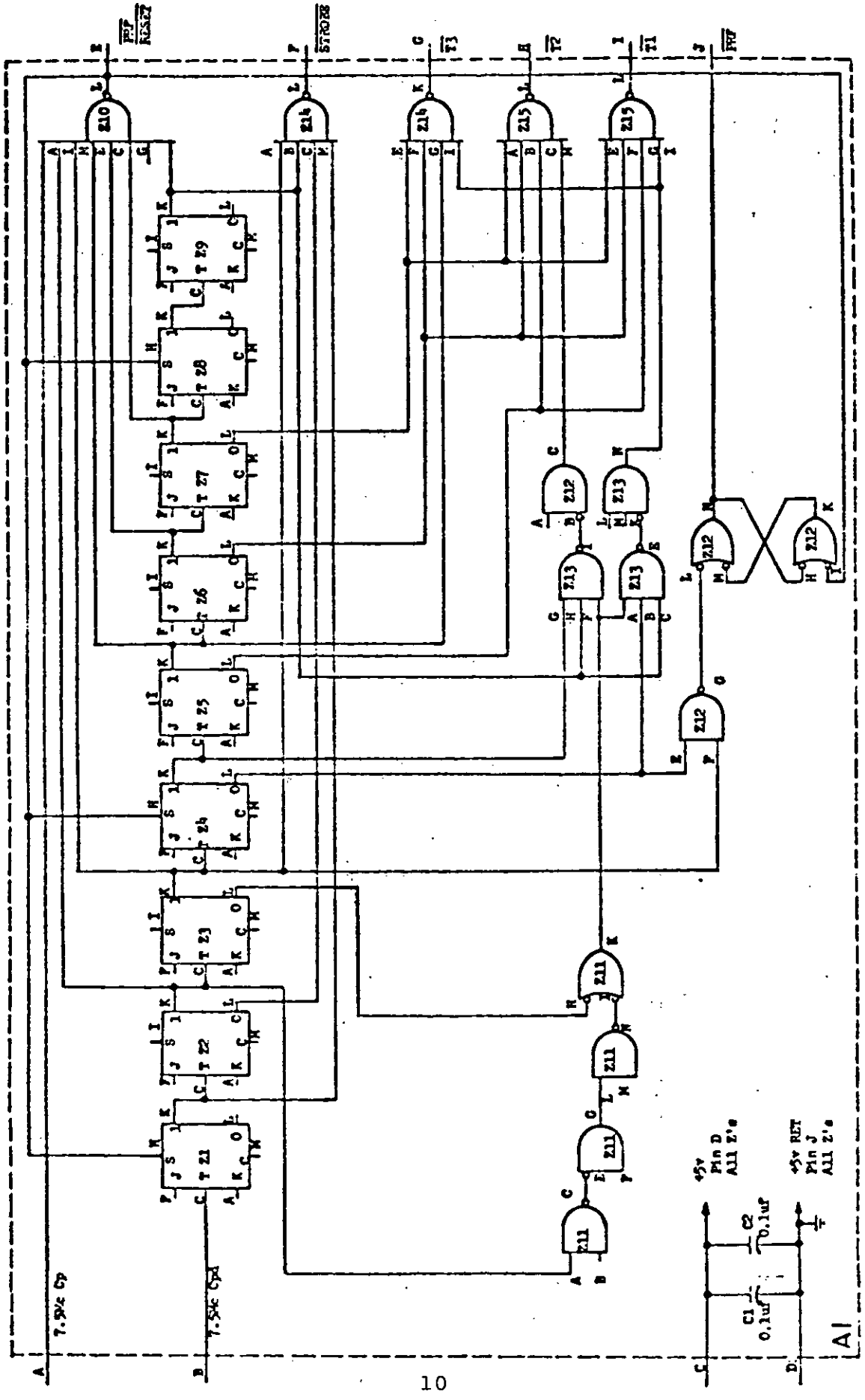


FIGURE 3. TEST FIXTURE SIGNAL GENERATOR

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TABLE IV. TEST FIXTURE SIGNAL GENERATOR PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
C1	Capacitor, 0.1 uF at 50 Vdc, SSM-1-189, Gulton Industries, Inc. or equivalent.
C2	Same as C1.
Z1 thru Z9	Integrated circuit, SF-50-28, Sylvania or equivalent.
Z10	Integrated circuit, SF-160-21, Sylvania or equivalent.
Z11	Integrated circuit, SF-140-37, Sylvania or equivalent.
Z12	Same as Z11.
Z13	Integrated circuit, SF-190-21, Sylvania or equivalent.
Z14	Integrated circuit, SF-40-28, Sylvania or equivalent.
Z15	Same as Z14.

4.6.3 Performance. The selected range counter circuit card assemblies shall be tested as follows: Set up equipment as shown in Figure 4. Use oscilloscope, item 3, for observation of input and output signals specified in 4.6.3.2 through 4.6.3.7. When specified, a logic 0 shall be observed as a 0.0 $\pm$ 0.5 Vdc level and a logic 1 as a +3.5 $\pm$ 1.0 Vdc level. Switches S1 through S10 on test fixture, item 4, are used to supply the logic level input signals specified in 4.6.3.4 through 4.6.3.7. Switches supply a logic 1 in position 1 and a logic 0 in position 2.

4.6.3.1 Input power. Using item 1, Table II, apply input power as specified in 3.5.1.

4.6.3.2 Clock signals. Using item 2, Table II, apply a 7.5 $\pm$ 0.5 megahertz (MHz) signal to J2 and with oscilloscope, item 3, Table II, observe clock output signals Cp and Cpd at TP1 and TP2 respectively to determine compliance with 3.5.2.

4.6.3.3 Input signals. Using item 3, Table II, verify input signals PRF $\overline{\text{RESET}}$, $\overline{\text{STROBE}}$, T3, T2, T1, $\overline{\text{T1}}$, and PRF at TP3 through TP9 respectively for compliance with Figure 1. The input signals are generated on test fixture subassembly A1, as shown in Figure 3.

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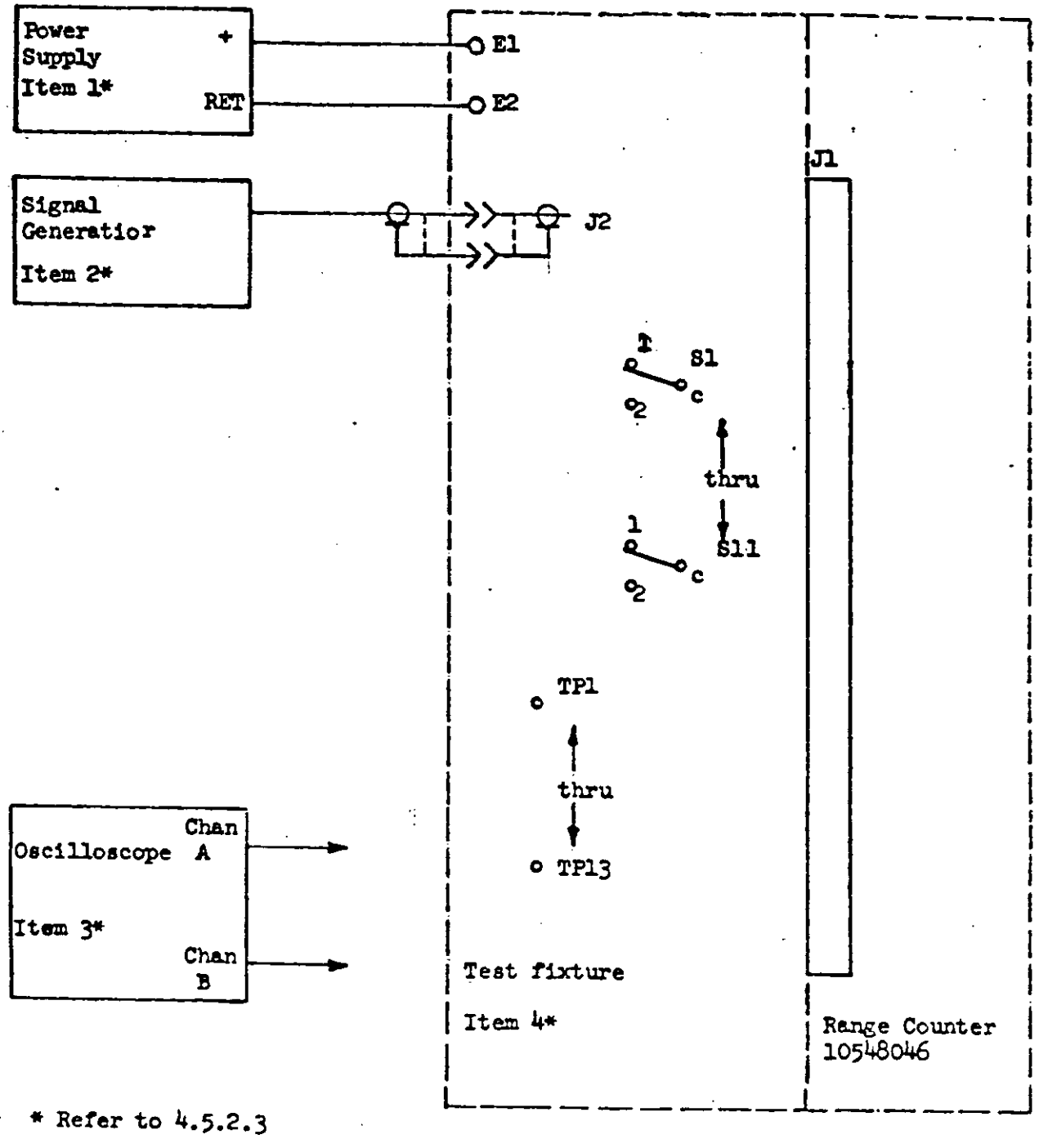


FIGURE 4. RANGE COUNTER TEST SETUP

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4.6.3.4 MIN RANGE INHIBIT output. Use switch S1 to apply logic levels as specified in 3.5.4. At TP10, observe the MIN RANGE INHIBIT output signal as specified in 3.5.4.

4.6.3.5 E/L COURSE output. Use switches S2 through S10 to apply logic levels, as specified in Table I, to terminals P1-12, P1-56, P1-21, P1-33, P1-35, P1-38, P1-40, P1-48, P1-55 respectively. Use switch S11 to apply logic levels as specified in 3.5.5 to P1-4. At TP11, observe the E/L COURSE output signal as specified in 3.5.5.

4.6.3.6 PRE RANGE TRIG output. Use switches S2 through S10 as specified in 4.6.3.5 to apply logic levels as specified in 3.5.6. At TP12, observe the PRE RANGE TRIG output signal as specified in 3.5.6.

4.6.3.7 T2 output. At TP13, observe the T2 output signal as specified in 3.5.7.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet PS10548046. The level of protection shall be as specified in the procurement document.

6. NOTES. As specified in 6. Notes of MIL-C-50652.

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3. DOCUMENT TITLE

CIRCUIT CARD ASSEMBLY: 10548046 (COUNTER, RANGE)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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